

An IoT Based Wild Animal Intrusion Detection System

PROJECT REPORT

Submitted by

ABIYA MARIYAM ALEX

PRC22CSOT003

M YADHU KRISHNA

PRC22CSOT016

NANDANA BIJU

PRC22CSOT017

SANIA SAJAN JOHN

PRC22CSOT022

To

APJ Abdul Kalam Technological University

*in partial fulfillment of the requirements for the award of B. Tech Degree in
Computer Science and Engineering*

(Internet of Things, Cyber Security including Blockchain Technology)



Department of Internet of Things & Cyber Security

Providence College of Engineering, Chengannur

March 2025

DEPARTMENT OF INTERNET OF THINGS & CYBER SECURITY
PROVIDENCE COLLEGE OF ENGINEERING, CHENGANNUR



CERTIFICATE

Certified that this report entitled '*An IoT Based Wild Animal Intrusion Detection System*' is the report of project completed by the following students during **2024-2025** in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in Computer Science and Engineering (Internet of Things, Cyber Security including Blockchain Technology).

ABIYA MARIYAM ALEX

PRC22CSOT003

M YADHU KRISHNA

PRC22CSOT016

NANDANA BIJU

PRC22CSOT017

SANIA SAJAN JOHN

PRC22CSOT022

Ms. Aswathy Krishnan B (Project Supervisor)

Assistant Professor, Department of Internet of Things & Cyber Security (CSOT)

Ms. Praseetha S Nair (Project Co-Supervisor)

Assistant Professor, Department of Internet of Things & Cyber Security (CSOT)

Dr. Pramod Mathew Jacob (Mini Project Co-Ordinator)

Associate Professor & HoD - CSOT

Department of Internet of Things & Cyber Security (CSOT)

Providence College of Engineering, Chengannur

DECLARATION

We, hereby declare that, this project report entitled '*An IoT Based Wild Animal Intrusion Detection System*' is the bonafide work of ours carried out under the supervision of **Ms. Aswathy Krishnan B**, Assistant Professor, Department of Internet of Things & Cyber Security. We declare that, to the best of our knowledge, the work reported herein does not form part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion to any other candidate. The content of this report is not being presented by any other student to this or any other University for the award of a degree.

Sl. No.	Name of Student	Roll Number	Signature
1	ABIYA MARIYAM ALEX	PRC22CSOT003	
2	M YADHU KRISHNA	PRC22CSOT016	
3	NANDANA BIJU	PRC22CSOT017	
4	SANIA SAJAN JOHN	PRC22CSOT022	

Ms. Aswathy Krishnan B (Project Supervisor)

Dr. Pramod Mathew Jacob

Head of the Department - Computer Science and Engineering

Providence College of Engineering

Date: 31/03/2025

ACKNOWLEDGEMENTS

We take this opportunity to express our deep sense of gratitude and sincere thanks to all who helped us to complete this project successfully.

We are deeply indebted to our Project Supervisor **Ms. Aswathy Krishnan B**, Associate Professor for her excellent guidance, positive criticism, and valuable comments. We express our sincere gratitude to our project coordinator **Dr. Pramod Mathew Jacob**, Associate Professor for his timely guidelines and critical reviews. We are thankful to our co-supervisor **Ms. Praseetha S Nair**, Assistant Professor for her constant support and guidance.

We are also greatly thankful to our Head of Department **Dr. Pramod Mathew Jacob**, Associate Professor for his continuous support. We also thank the other department faculty members who helped us in doing this project work.

Finally, we thank our parents, family members and friends who directly and indirectly contributed to the successful completion of our main project.

Abiya Mariyam Alex

M Yadhu Krishna

Nandana Biju

Sania Sajan John

Date: 31/03/2025

ABSTRACT

The human-animal conflict has been increasing in Kerala for the last 10 years, resulting in deaths and injuries to farmers close to the forest areas and damage to crops. As per the local news articles and media outlets, a total of 1,300+ people have lost their lives to wild animal attacks since 2011. Recently, the number of attacks by elephants, tigers and wild boars has been increasing in the state. The lack of provisions for compensating for the loss of lives and crop damages is haunting the farmers, forcing them to withdraw from farming. This project is to bring a scientific solution to the unending menace faced by the farmers. This project is introduced to solve the conflict between the human and the wild animals in a peaceful way by using the deterring methods like quick flashes of lights and high frequency loud noises. This project uses a custom ML model built using TensorFlow Lite which is used for building ML models for edge devices. Raspberry pi is used in the project which is a low power computer. Given the rise in attacks on Keralan agriculture, the goal of this project is to deter wild animals from targeting agricultural regions. This project, which is built on the Raspberry Pi platform, identifies wild animals using machine learning. Along with observing and monitoring the real-time broadcast from the Raspberry Pi, the system also includes a companion Android app that enables the user to operate the lights in the field. The deterrence mechanism can be activated at the user's choice because of the manual control.

Keywords: Machine learning (ML), TensorFlow Lite, Raspberry Pi, Android app, Animal identification, Deterrence mechanism

TABLE OF CONTENTS

ACKNOWLEDGEMENTS	i
ABSTRACT.....	ii
LIST OF FIGURES.....	vi
LIST OF TABLES.....	vii
LIST OF ABBREVIATIONS.....	viii
CHAPTER 1.....	1
INTRODUCTION.....	1
1.1 Background	1
1.2 Existing System	1
1.3 Problem Statement	2
1.4 Objectives.....	2
1.5 Scope.....	3
CHAPTER 2.....	4
LITERATURE REVIEW	4
CHAPTER 3.....	13
SYSTEM ANALYSIS.....	13
3.1 Functional Requirements.....	13
3.2 Non-Functional Requirements.....	15
3.3 Feasibility Analysis.....	15
3.3.1 Technical Feasibility.....	15
3.3.2 Operational Feasibility.....	16
3.3.3 Economic Feasibility.....	16
3.3.4 Behavioral Feasibility.....	16
3.4 Hardware Requirements.....	16
3.5 Software Requirements	17
3.6 Life Cycle Used	17

3.7 Software Cost Estimation	18
3.8 Hardware Cost Estimation	19
3.9 Total Product Cost Estimation	19
3.10 Project Scheduling using Gantt chart.....	20
CHAPTER 4	21
METHODOLOGY	21
4.1 Proposed System.....	21
4.1.1 Animal Detection.....	22
4.1.2 Animal Deterrence.....	23
4.2 Image Dataset.....	23
4.2.1 Training.....	24
4.3 Device Management.....	25
4.4 Advantages of Proposed System.....	25
CHAPTER 5	26
SYSTEM DESIGN	26
5.1 Architecture Diagram.....	26
5.2 Circuit Diagram.....	27
5.3 Flow Chart.....	28
5.4 Use Case Diagram.....	29
5.5 Activity Diagram.....	30
5.6 Sequence Diagram.....	31
5.7 Collaboration Diagram.....	32
CHAPTER 6	34
SYSTEM IMPLEMENTATION	34
6.1 Machine Learning Model.....	34
6.2 Implementing the Model into Android App.....	35
6.3 Hardware Implementation.....	36
6.3.1 Raspberry PI 5.....	36
6.3.2 ESP 32.....	36
6.3.3 SIM 800L.....	37
6.3.4 Relay Module.....	38
6.3.5 Web Camera.....	38

6.3.6 Buzzer.....	39
6.3.7 LED Light.....	39
6.3.8 Battery.....	40
CHAPTER 7.....	41
TESTING.....	41
7.1 Testing.....	41
7.1.1 Types of Testing.....	41
7.1.2 Test Summary Report.....	42
CHAPTER 8.....	43
RESULT.....	43
8.1 Software Module.....	43
8.2 Hardware Module.....	46
CHAPTER 9.....	47
CONCLUSION.....	47
REFERENCES	48
APPENDIX 1: SAMPLE CODE.....	50
APPENDIX 2: BILLS OF PURCHASED MATERIALS.....	55

LIST OF FIGURES

Figure number	Figure Name	Page Number
Figure 3.1	Incremental Model	17
Figure 3.2	MDLC Model	18
Figure 3.3	Project Schedule using Gantt chart	20
Figure 4.1	Flowchart of the system	21
Figure 4.2	Examples for animals that are detected at night	22
Figure 4.3	Wild Animal Images	24
Figure 5.1	Architecture diagram of Animal Detection System	26
Figure 5.2	Circuit Diagram	27
Figure 5.3	Flow Chart of Animal Intrusion Detection System	28
Figure 5.4	Use Case Diagram of the system	29
Figure 5.5	Activity Diagram of the system	30
Figure 5.6	Sequence Diagram of the system	31
Figure 5.7	Collaboration Diagram of the system	32
Figure 6.1	Raspberry Pi 5	36
Figure 6.2	ESP 32	37
Figure 6.3	SIM 800L	37
Figure 6.4	Relay Module	38
Figure 6.5	Web Camera	38
Figure 6.6	Buzzer	39
Figure 6.7	LED Light	39
Figure 6.8	Battery	40
Figure 8.1	Start Screen	43
Figure 8.2	Detecting Wild Animals	45
Figure 8.3	Animal Detection System	46

LIST OF TABLES

Table Number	Table Name	Page Number
Table 3.1	COCOMO Model Coefficients	18
Table 3.2	Hardware Cost Estimation	19
Table 7.1	Decision Based Testing	39
Table 7.2	Testing Summary of IoT Module	39

LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CCTV	Closed-Circuit Television
CNN	Convolutional Neural Network
COCOMO	Constructive Cost Model
GPU	Graphics Processing Unit
GSM	Global System for Mobile Communications
HD	High Definition
IDE	Integrated Development Environment
IoT	Internet of Things
IR	Infrared
KLOC	Kilo Lines of Code
LOC	Lines of Code
MDLC	Model Development Life Cycle
ML	Machine Learning
MQTT	Message Queuing Telemetry Transport
NFR	Non-Functional Requirement
PIR	Passive Infrared
RTC	Real-Time Clock
SDLC	Software Development Life Cycle
SMS	Short Message Service
SSD	Solid-State Drive
UML	Unified Modeling Language
Wi-Fi	Wireless Fidelity
WSN	Wireless Sensor Network
RTDB	Real-Time Database

CHAPTER 1

INTRODUCTION

1.1 Background

Kerala has been seeing a rise in wild animal attacks on farmlands, this project aims to distract these animals from the fields of agriculture. This system is based on Raspberry Pi platform and uses machine learning to identify wild animals. The system additionally features an Android app enabling users to view and access the live feed from the Raspberry Pi, as well as they can manually control the lights in the field and trigger loud noises to drive away the wild animals.

Since the catastrophic floods of 2018 in Kerala led to extensive soil erosion, loss of plantation and alterations in water routes. As a result, this increased invading of animals like elephants, wild pigs to move into the places where humans are settled, escalating conflict between animals and humans in that particular region. Due to their natural habitats being destroyed they were deprived of their resources which led them to raid fields of agriculture and livestock. The ongoing conflicts between humans and wildlife has led to increase in human attacks with the death of wild animals.

This work seeks to resolve this problem peacefully by distracting the animals with sudden light flashes and high frequency, loud sounds. This system utilizes a model built with TensorFlow Lite, a mobile library for building ML models for edge devices. It is an ideal choice, given that the Raspberry Pi is used in the system.

1.2 Existing System

The existing system before implementation of animal intrusion detection system there were traditional methods used to manage and prevent wild animal intrusion in agricultural areas. These methods were physical barriers, manual monitoring, scare tactics, guard animals and traditional traps.

Physical barriers: Fencing or walls to keep animals out of crops or livestock areas.

Manual monitoring: Farmers or security personnel would regularly patrol fields to check for signs of animal intrusion and take action as needed.

Scare tactics: Use of loud noises, firecrackers, or visual deterrents to frighten animals away from crops or settlements.

Guard Animals: Some farmers used dogs or other animals trained to protect crops and livestock from wild animal attacks.

Traditional traps: Setting up physical traps to catch wild animals intruding on the property

However, the above methods were proven to be ineffective since it required a lot of efforts from the farmers as most of these require human intervention, it made them ineffective during nighttime hours or in remote areas where monitoring was difficult.

Many traditional deterrents, such as loud noises or visual scare tactics were initially effective but became less useful over time as animals adapted to the stimuli. Manual monitoring and even using guard animals could be risky, as encounters with wild animals could result in injury or death to both humans and livestock.

1.3 Problem Statement

To develop an automated wild animal detection system for preventing animal intrusion into agricultural fields using TensorFlow Lite.

1.4 Objectives

The following are some of the goals of our proposed paradigm:

- To keep animals away from the farmlands by avoiding harm to the wildlife and the landowner.
- To use a Raspberry Pi based system with a camera and a machine learning model to identify animals.
- To develop an Android application for monitoring the live feed, control lights and play sounds to manually deter the animals.

- To use TensorFlow Lite, a mobile library for building ML models for edge devices, ensuring the system is suitable for the low-power Raspberry Pi.

1.5 Scope

The system targets to eradicate the human-animal interaction peacefully by scaring animals using sudden flashes or triggering unpleasant sounds with high-frequency. The scope of the work is to detect the animals that create a mess on the farmlands and distract them automatically by not involving the landowner or the device user. The intruder detecting module running in the system will detect intruder and activate the prevention mechanism. Additionally, the scope of the project is restricted to landowners of agricultural, the system applicability extends to homes and neighborhoods where the intrusion of wild animal intrusion is common. This project automatically identifies the wild animal that is intruding by activating all the mechanisms to drive the animal away, without requiring the direct involvement of the owner.

CHAPTER 2

LITERATURE REVIEW

We have analyzed various existing work in the field of wild animal intrusion system using various components. The summary of the most relevant 12 papers is provided below.

Jinpeng et al. has proposed a model [1] for an Intelligent Farming System Using Fog Computing for Wildlife Detection. This paper proposes a smart agriculture infrastructure using LoRa-enabled fog computing to overcome the limitations of cloud-based infrastructure. They used Passive Infrared sensors, and a 360-degree camera which is attached to the Raspberry Pi. The system is structured into two interconnected tiers: a Sensing Layer for data collection and a Fog Computing Layer for data processing and analysis, which exchange information through reciprocal communication channels. In Sensing Layer, the sensors spread across the farm measure things like temperature, humidity, and soil conditions. In Fog Layer one or more servers' control and manage all the sensors and devices on the farm. The fog nodes execute animal intrusion detection. There are three micro services. A camera is connected to Raspberry Pi which takes the pictures of the wild animals then identifies them through computer vision algorithms. Data collected from the sensors is sent to cloud via internet, where it's stored, analyzed, and used to inform decisions. This system detects animals before they enter the farm, identifies them, predicts where they'll go next, and quickly alerts farmers. To resolve key challenges, these innovative approaches focus on minimizing delays, reducing data transmission costs, strengthening rural connectivity, and maintaining affordability for agricultural stakeholders. They have implemented this model in agricultural areas where the likelihood of wild animals is more, this is mostly the rural areas. Their system uses LoRa technology to overcome internet connectivity issues and slow data processing. It also features a flexible architecture for real-time services, multiple sensor arrangements, and a smart algorithm that tracks animal movements without harming them. The system faces challenges when animals are too close to camera, as accuracy decreases. Animals which move fast also pose a problem, as they quickly move out of range or get close, which makes hard to predict their next location. This can lead to

cameras capturing the wrong area. Additionally, small animals that are hard to capture clearly can be difficult to detect. Cameras must be positioned to capture animals within their field of view.

Jeevitha & Vengatesh has proposed a model [2] for A Study on Animal Alert System Using Sensors and Camera Technology. They developed a system that uses wireless sensors to automatically send alert messages with images to landowners and forest officials when animals intrude, using mobile network technology. When the animal moves, the sensor will detect it, the camera will take a picture, and a microcontroller will classify the picture. Following signal transmission to the camera, the microcontroller receives the image then forwards it to the module which processes the image, where three-level segmentation is used to process the image's objects. These techniques and methods are used to scare animals and to avoid the resulting dangers to human and the loss of resources. They have implemented this model in agricultural fields, rural areas and areas where the likelihood of animal intrusion is more. Their model has an alarming message that makes a sound which scare the animal without harming it. The sensors used are wireless. CCTV surveillance cameras or built-in sensor-based cameras are used to record live video. Additionally, it is affordable, sturdy, dependable, and simple for farmers to obtain. However, there are drawbacks to their project as well. For example, using a camera to detect animals involves more image processing in a live video, and any wild animals that move more quickly will be taken as blurry images, which could result in misclassification because the system won't be able to detect an attack.

Bharathan & Suganthi has proposed a model [3] using Internet of Things (IoT) to detect and prevent animal damage in agricultural fields. To identify animals on agricultural land, they have employed the YOLOv3 object detection technique, which is based on a deep learning model driven by machine learning. When the animals are detected the cameras and sensors send a notification to the farmers. It also makes noises to scare the animals, this noise will be different for each species. The extent of damage the animal does is directly connected with the noise level. A sensor on a Raspberry Pi uses pre-loaded big datasets to identify the animal. A distinct sound is then generated, and a distinct order is transmitted. The farmer's smartphone app or remote control can be used to quiet the buzzer. These techniques are used to resolve the issues such as the harm

caused by animals in agricultural fields and to humans. They have implemented this model in farmlands and rural areas where there are more wild animals. Their model shows a mechanism to scare off animals from agricultural fields without causing harm to both humans and the animal. The reason YOLO's method is effective is that it anticipates which regions of the image will have the required data and only uses the CNN classifiers to process those regions. This custom CNN processes each image data quickly. The farmer is notified through WhatsApp. One of the project's concerns is that the detection success rate drops when a wild animal is quite near to the camera. The technology is challenged by fast-moving animals since it is hard to forecast where they will go next because they can either leave the view of the camera or approach too closely. Small animals that the camera cannot record well are also hard to find. The system requires regular maintenance to ensure that the cameras, sensors, and other equipment are functioning correctly. Scaling the system to cover larger areas or more animals can be a challenge.

Meenakshi et al. has proposed a model [4] for Animal Intrusion Detection and Ranging system using YOLOv4 and LoRa. This suggested system has been implemented using two systems: the receive communication system and the detector and transmitter system. A Raspberry pi, a LoRa transmitter, and a visual surveillance system make up this detection system. A microprocessor, microcontroller, or receiver LoRa compose the receiving communication. A camera that is wirelessly connected via a CCTV or live webcam, or one that is physically attached to the Pi, can serve as the image monitoring device. The output that detected is acquired after the input image has been processed and analysed by the image detection module. The image detector is trained by the data consisting of wild animals and domestic animals. After being taken by the camera and analyzed using a pre-trained dataset, the image is classified whether an animal is identified. The receiver communication module then receives the classified animal data. The recipient of the communication is linked to a network device is based on WiFi and has network access. This makes it possible to use the API to create Web applications, Android apps, and SMS-based communication. This is the point at which the identified information is used to communicate with the right person depending on whether the animal being categorized is a domestic or wild animal. Creating a sustainable environment for resource sharing between humans and animals is the primary goal of this article and the suggested method. Additionally, it's to prevent damage from wild

animals intruding. The suggested method eliminates friction between humans and animals without endangering either. According to their model, it has an Android application that shows the animal that has been spotted. If smartphones are not available, the same API is used to send the relevant user a text message. The system can be set up in a variety of places, including farmlands, hilly areas, and forests. When a detection occurs in a location, the system notifies the officials of the projected range for that region. This helps to provide the best possible response and to find and protect the farmers and animals in terms of property and life as soon as possible. Their strategy, however, ignores the reality that the system just sends out warnings and does not actively prevent animals from entering fields or other locations; instead, it leaves it up to the authorities or farmers to take the necessary measures. YOLOv4's inaccurate object classification could result in missed detections or false alarms. The efficiency of the gadget may be diminished by low light levels, thick foliage, or extreme weather. Setting up a LoRa network in a remote or inaccessible area can be costly and logistically challenging. Sensors and cameras need a steady power supply. In remote locations, the receiver communication system's Wi-Fi-based network, which it uses to relay messages, may not always be accessible.

Kathir et al. has proposed a model [5] for Animal Intrusion Detection Using Yolo V8. The data for this work was collected for training an embedded device to perform real-time animal detection. This data contains at least 376 images for each animal collected through various browsers. YOLO v8, which is the most recent iteration of Ultralytics popular YOLO series of models, which are renowned for their speed and accuracy, has been chosen. The workflow in this project is Data Pre-Processing, Creating Working Directories, Extracting and Visualizing Test Images, Separating Training and Validation Dataset, Image Visualization with Bounding Box, Model Training. The primary experiments in this study were conducted using the Ultralytics framework in conjunction with Python. Local experiments were executed on a GTX 1050 GPU. It addresses the problem of wildlife intrusions in agriculture as its main goal, the presented work focuses on precise animal identification and the activation of appropriate warning systems. This technology effectively identifies animals in agricultural fields, protects the crops and also make sure of the welfare of wildlife. They have implemented this monitoring system that can detect the image of an animal entering, sending alert messages to the farmers or landlords. The advantages of this

work can be YOLO v8 object detection for an Animal Intrusion Detection System, effectively safeguarding agricultural resources and managing human-wildlife conflicts. It can overcome real-time performance challenges, ensuring system reliability with a user-friendly interface. The implementation demonstrates advanced technology's efficacy in solving practical challenges, fostering harmony between agriculture and the environment for future resource preservation. The flaws in this system can be the system's performance may degrade under certain weather conditions such as heavy rain, fog, or low light, which could affect the accurate working of the YOLO v8 object. Farmers or landlords may require technical knowledge or support to install, configure, and maintain the system. In remote places this system can be difficult to maintain.

Anuvind et al. has proposed a model [6] for Development and Implementation of an animal intrusion detection system using image and audio processing. Deep learning, image processor, audio processor, and the YOLO system have all been applied. The animal is recognized and located using image processing. An alarm notification is sent to the appropriate authorities and the image is stored in cloud storage when the animal is identified. Both animal species and the time of detection are included in the alert message. The microphone does the same, recording the sounds around it and feeding them into the sound model. The relevant authorities receive a warning notice whenever an animal sound is detected. To detect the species of animals that invade human environments, the system's picture and audio processing components collaborate. This is utilized to address the growing conflict between humans and animals in modern times. A major issue that endangers human life and damages resources is the existence of wild animals in densely inhabited regions. The damage caused by wild animals entering human areas can be lessened by using an animal incursion detector system that processes audio and images. We can stop property damage, injuries, and even fatalities by identifying the entry early on. They have put in place a monitoring system that can identify the sound and visual signs of an animal entering and notify the appropriate authorities. One of the project's benefits is that the suggested animal infiltration system can identify different kinds of animals. The damage produced by wild animals entering human areas can be lessened with the use of an animal intrusion detector system that has both picture and sound detection capabilities. In addition to helping to preserve crops and guarantee home safety, this technology provides a solution to disputes between humans and wildlife. The disadvantages in this technology can be continuous

operation of cameras, microphones, and other system components may require a stable power source, which is difficult in remote places. The system only sends alerts and does not actively prevent animals from entering fields or areas, leaving the responsibility of action entirely on the authorities or farmers. The system uses cameras and microphones as the primary detection tools which may fail in environments with dense vegetation, extreme weather, or poor lighting conditions.

Prabhat et. al. has proposed a model [7] for Implementation of Wild Animal Intrusion Detection Model Based on Internet of Things. They have used E-Vehicle integrated with an ESP32 camera module to get the images on invading animals, IR sensors, which help the robot follow a prescribed path to assure it operates within defined areas, and RTC module to control the functioning of the vehicle for some definite time, thereby increasing the efficiency of operation for better efficiency in the whole system by automating in monitoring so that less human presence is required while the monitoring takes place]. Ultrasonic sensors surround the field to detect intruder movement. The IoT application sends an alert notification to the farmer when an intrusion has happened, and a buzzer sounds to keep animals off the land. Wireless transmission foresees the employment of the NRF24L01 module in the transfer of data between components, ensuring a seamless operation and connection as a means of seeing that the loss of immense resources is mitigated against as a courtesy to providing human safety. Mainly agriculture was taken into consideration while implementing this model. The integration of IoT technology sets the model in the smart farming regime while the undertakings in robotics associated with the model place it in technology and robotics. Their model proves to better protect agriculture by monitoring the agricultural fields against wild animal intrusions, its real-time monitoring capabilities thanks to IoT technology enable timely detection of any intrusions, using sensors, cameras, and robots illustrates innovation in agricultural practices aligned with smart farming principles, offering an affordable solution across wide agricultural operations with reduced reliance on traditional protective measures. However, high initial costs are likely to displace several farmers with limited finances. The batteries used in E-vehicles have a finite lifespan, which, along with the vehicles' subsequent replacement cost, adds to the long-term operational costs. Though E-vehicles generally require less maintenance compared to traditional vehicles, a challenge will present itself in rural areas, where specialized knowledge may be required for repairs. The system can

produce false alarms resulting from your harmless animal or the surroundings, causing unnecessary disturbances and alert fatigue. Users may face challenges operating and maintaining the system, primarily because they cannot operate and maintain the equipment without training and support.

Muneera et. al. has proposed a model [8] for a WSN and IoT-based Wild Animal Infringement Identification, Diversion and Alert System. Technologies like WSN and IoT have been used as a solution by them. A closed area system is developed to detect and report animal intrusion and to implement preventive measures in an automated manner. For animal detection, WSN nodes are placed at every corner of the field and equipped with a perimetric laser sensor. The central base station is informed of the intruder entry that activates all the nodes around the animal, which identifies the animal's presence, with the help of flashes and buzzers it will divert the animals from the field. Also, they set up a space with buffer crops that will provide food for wild animals, encouraging them to stay in that area. This helps maintain a balance in the ecosystem and prevents wild animals from going hungry. Messages are sent through the MQTT protocol which needs only smaller code footprints, which will be an advantage for the people in remote areas for faster communication. It helps in resolving the increasing conflict between humans and wildlife and significant damage caused to crops by wild animals leads to economic losses for farmers. They have implemented this model in agriculture focusing on crop protection and wildlife management. Their model shows effective crop protection without causing harm to the animals. It uses methods like buzzers and flashlights to deter animals. However, their model fails to address the false alarms by the use of sensors and detectors causing unnecessary alerts to the forest officers or farmers.

Manikandan et. al. has proposed a model [9] for IoT based animal detection and alert system for farm fields. The Arduino UNO microcontroller acts as a hub for all inter-device communication. In the farm field ultrasonic sensors and PIR sensors are used for detecting the animal presence. Using GSM module enables quick notifications to farmers via SMS while uses WI-FI for transmitting data and images. By camera integration and YOLO algorithm helps in capturing real-time images of detected animals and object detection in images. ThingSpeak and Adafruit platforms are used for data storage, processing and for displaying the information about animal detection

for resolving the issue of wild animal entering farm fields and destroying crops and human-wildlife conflicts. They have implemented this model in the domain of smart agriculture focusing on farm protection against wild animal intrusions. Their model shows multi layered deterrent mechanisms such as sound-based, visual-based, and environmental deterrents such as fog to prevent entry of wild animals. SMS notifications are sent via GSM technology and realtime pictures of the animals detected will be sent through a telegram bot. Automated and manual modes are supported in the control mechanism, the farmer can control the system using a remote manually.

Jeevitha and Vengatesh has proposed a model [10] for the Animal Intrusion Warning System. They have used image analysis methods for tackling issues in the agricultural sector that endanger human life and cause enormous resource losses. They have implemented this model in agriculture fields by for using a photograph to automatically notify the property owner and forest managers. When the animal moves, the sensor will detect it, the camera will take a picture, a microcontroller will classify the picture, and a GSM module will send the property owner and the forest department an SMS alert. Three steps are involved in the identification of animal intrusions: sensor and camera, microcontroller GSM module, and alert SMS messaging. Using a microcontroller, the sensor at the first level will identify how the animal is moving followed by transmitting signal to the camera. The following step involves the microcontroller receiving the taken copy and sending to a module where it processes the image, where three-level division, extraction of features, and groupings are used to process the captured objects. The GSM module receives signals during classification if the object image matches the database's stored image exactly, and the officials or landowner receive an alarm message. Ultrasonic, thermal, and passive infrared (PIR) sensors are the three different sensors put to identify animals. This model demonstrates how it successfully extends power supply service times and making it Cost-effective making it easily accessible for farmers. But their models fail to address the problems with a short battery life and components within the sensor that are incompatible, which causes challenges while uploading larger video files.

Jinpeg et al has proposed a model [11] for intrusion of animals which is a major concern of farmers in agricultural fields. They have used computer vision, cameras, and inexpensive PIR sensors to forecast animal intrusion to keep it low cost for farmers and

to resolve the internet disconnections in rural areas, also it eradicates the high speed and use of bandwidth which is related with cloud computing. They have implemented this model in smart agriculture. This type of agriculture combines latest technology, which combines WIFI, cloud computing and IoT technologies, relies upon a number of sensing nodes to achieve perfect tracking, surveillance, automating, and examining activities with two layers sensing layer and fog computing layer. Since they have implemented LoRa and Lo-RaWAN facilitated framework, which supports minimum power, extensive networking intended to connect energy-constrained devices wirelessly to a remote server within 2 km radius and the problem of inadequate internet access has been addressed. A Raspberry Pi is used to capture that will be analyzed to identify animals, and one or more servers make up the fog layer, which offers administrative management over the complete IoT infrastructure of agricultural land. Their model shows how the power is efficiently managed making the system energy efficient which also reduces the dependency on internet connection. But their model fails to address how it may not be suitable for vast farm areas where cameras cannot cover the entire surface area, this can also lead to triggering of false positives in response to environmental factors like wind or rain.

Praveen and Divya [12] have proposed a demonstrate for animal interruption discovery in rural areas for early location of interruption around the field by capturing the photo of intruder and classifying them which empowers us to take appropriate activities based on the sort of the intruder. They equipment utilized in this framework are PIR HC-SR501, webcam, microcontroller shining light emitter and sound generator and the program utilized in this framework are Arduino IDE for programming the microcontroller and MATLAB 2017a for picture handling and classification. They have implemented this model in agribusiness and natural life administration, particularly centering on checking agrarian areas to avoid wild creature interruptions that can lead to clashes and asset misfortune. Their model shows how to secure crops and animals from wild creatures and sends quick cautions to partners, empowering speedy activity. But their model fails to distinguish certain animals or misclassifying them, which can lead to potential dangers.

From our analysis, we found out that the system just sends out warnings and does not actively prevent animals from entering fields or other locations.

CHAPTER 3

SYSTEM ANALYSIS

The system analysis phase includes the analysis of various functional requirements, non-functional requirements, design constraints and hardware requirements. This chapter provides the detailed system analysis of animal intrusion detection system.

3.1 Functional Requirements

The proposed Animal Intrusion Detection System is expected to meet the following functional requirements.

FR1: Video Recording Functionality

Description: The system should record the live footage of wild animals approaching the agricultural field using web camera (HD 1080p) and ML modules.

Input: Live video feed from the camera.

Output: Video of the detected animal in MP4 format.

FR2: Machine learning model for identification

Description: The system should use custom-built machine learning model (TensorFlow Lite) to classify animals into distinct categories based on the camera feed.

Input: Identified animal class from the machine learning model.

Output: Triggered audio and light signals, no file output (real-time action).

FR3: Deterrent Activation

Description: Once the animal is detected, the system must trigger deterrent mechanisms such as loud sounds and flashlights. This action should be immediate.

Input: Identified animal class from the machine learning model.

Output: Triggered audio and light signals, no file output (real-time action).

FR4: Remote monitoring

Description: User should be able to monitor system performance and animal detections remotely through the companion app.

Input: Real time streaming from system sensors and camera.

Output: Real time update on animal detection and system status updated on companion app.

FR5: Energy efficient operation

Description: The system should operate efficiently using minimal power consumption, using low-power devices like the Raspberry Pi.

Input: Power supply data, such as voltage and current readings, in CSV format.

Output: Systems status in terms of power usage.

FR6: Data storage and backup

Description: The system should store animal detection data and videos securely for later review, either locally or on the cloud.

Input: Video footage in MP4 format.

Output: Stored video files or log data in cloud or local storage in MP4 format.

FR7: System calibration and updates

Description: The system should allow for regular updates and calibration of the machine learning model to improve accuracy over time.

Input: New training data in JPEG/PNG format.

Output: Updated model files and logs in structured format.

FR8: Encryption of data

Description: The system should protect all data's transmitted between sensors, camera, hubs and cloud storage to protect sensitive information and ensure privacy. This is especially important for preventing unauthorized access to animal detection data and user alerts.

Input: Data from camera feed (live video), animal detection data and user alerts. This data will be encrypted before transmission to any cloud service or between devices.

Output: Encrypted data that ensures confidentiality and integrity during transmission

3.2 Non-Functional Requirements

The proposed Animal Intrusion Detection System is expected to meet the following non-functional requirements.

NFR 1: Availability: The system should be available for the user 24×7 without failure.

NFR 2: Reliability: The system should be reliable with failure free operations. The reliability value ranges from 0 to 1.

NFR 3: Maintainability: The system should be maintainable without changing the base structure.

NFR 4: Security: The system should not be vulnerable to the attackers. The data gathered and stored by the system should be secure.

3.3 Feasibility Analysis

The feasibility study for the above-mentioned requirements is done and it is concluded that it is practically possible to build such a system. The technical, economic, operational and behavioral feasibility analysis is discussed below.

3.3.1 Technical feasibility

Both the Raspberry Pi 5 and the ESP32 are popular microcontroller systems because they have a lot of processing power and communication possibilities. A lightweight version of TensorFlow designed for edge devices, TensorFlow Lite is ideal for executing machine learning models on platforms with limited resources, such as the Raspberry Pi and ESP32. Integrating the hardware components with the machine learning model and developing the necessary software for controlling sound and light emissions is technically feasible. Ensure that the power consumption of the system components is within acceptable limits. TensorFlow Lite, a condensed version of TensorFlow made for edge devices, is perfect for running machine learning models on low-resource platforms. Due to its high processing power and connectivity capabilities, the Raspberry Pi 5 and the ESP32 are both well-liked microcontroller systems.

3.3.2 Operational feasibility

It should be easy for farmers to install, set up, and utilize the system and related app because to its user-friendly design. To avoid agricultural damage, the system must be dependable in identifying and successfully discouraging wild animals. To ensure the solution's long-term viability, take into account how simple it is to maintain, including hardware maintenance, software upgrades, and troubleshooting.

3.3.3 Economic feasibility

Compare the expected benefits such as less crop loss and higher agricultural productivity with the price of the hardware (Raspberry Pi, ESP32), sensors, development tools, and software licensing. To determine whether farmers in Kerala, India, and maybe other areas dealing with comparable issues with wild animal assaults on agricultural crops would want such a solution, do market research.

3.3.4 Behavioral Feasibility

Using surveys or interviews, find out how eager farmers are to use the system and the app, taking into account their opinions on technology adoption and wildlife control strategies. Design the companion app with the target customers, making it user-friendly. Farmers are instructed on how to use the system

3.4 Hardware Requirements

- Raspberry Pi
- Raspberry Pi case and cooler
- Web Camera HD 1080p
- Bread Board
- Male to female connector
- Male to male connector
- ESP32
- SIM800L 3.4V-4.4V
- LED Bulb 5V, 200-500 lumens
- Buzzer 3V
- Case for battery

3.5 Software Requirements

- Android
- Firebase
- TensorFlow Lite
- PI OS Bookworm

3.6 Life Cycle Used

In this project, we choose the incremental model. It is an iterative enhancement model. We develop our project as different modules which will be completed as different iterations. The Incremental model shown in Figure 3.1 is flexible and is easier to incorporate new features during the development phase

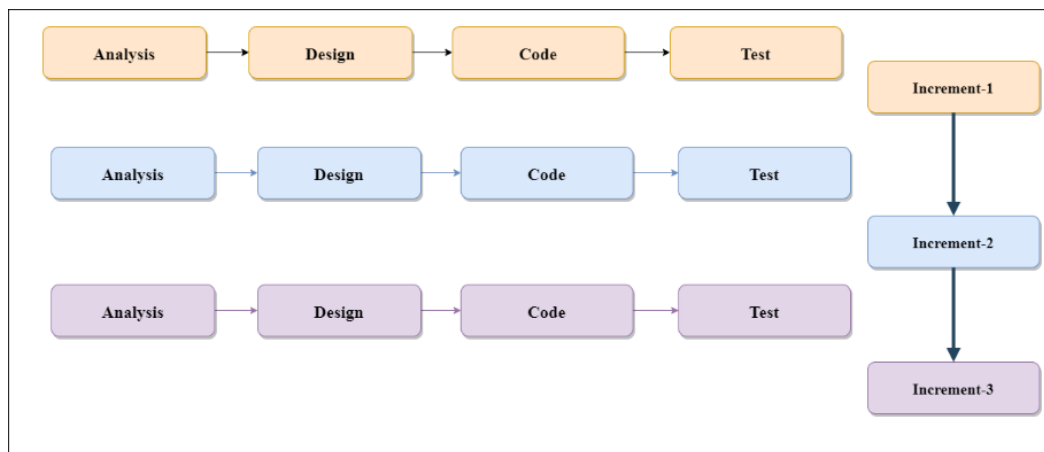


Figure 3.1: Incremental Model

In this project, we choose the incremental model. It is an iterative enhancement model. We develop our project as different modules which will be completed as different iterations. The Incremental model shown in Figure 3.1 is flexible and is easier to incorporate new features during the development phase. This approach allows for continuous improvements and adaptations as new requirements emerge. By breaking down the project into manageable iterations, we can better evaluate progress and make necessary adjustments at each stage. Analogous to the role of the software-development lifecycle (SDLC), the machine learning model-development lifecycle (MDLC) guides the activities of ML model development from inception through retirement. The key phases of the MDLC, including data ingestion, exploratory data analysis, model creation, and model operation, are illustrated in Figure 3.2. The MDLC ensures that the model is refined and optimized at every stage, enhancing both its accuracy and its

ability to meet the project's goals. It also provides a clear roadmap for addressing any challenges that may arise during the development process, ensuring smooth transitions between phases.

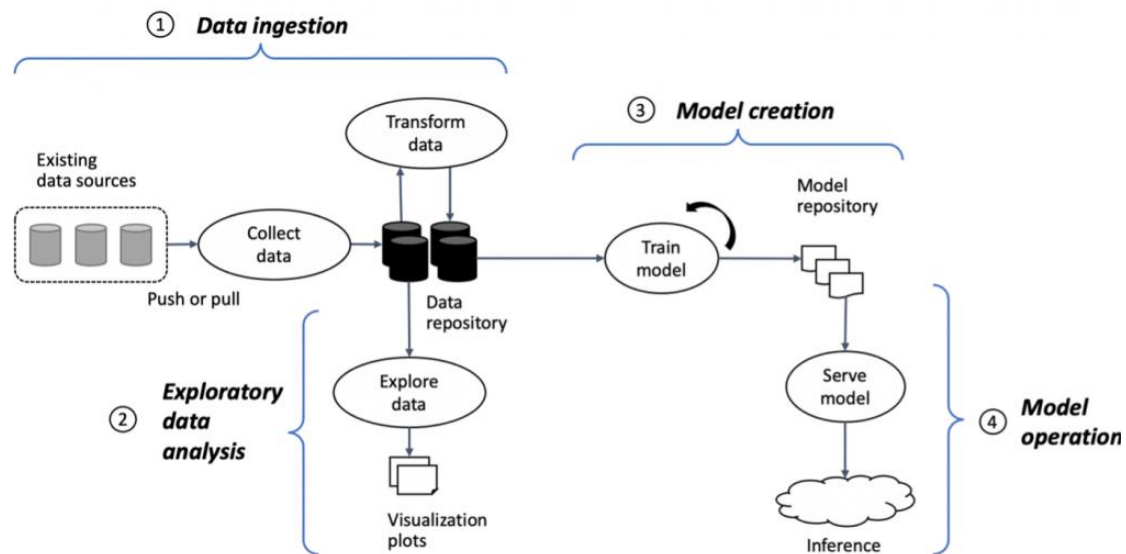


Figure 3.2: MDLC Model

3.7 Software Cost Estimation

For cost estimation, the COCOMO (Constructive Cost Estimation Model) is used.

Table 3.1: COCOMO model coefficients

Software Category	a	b	c	d
Organic	2.4	1.05	2.5	0.38
Semi-detached	3.0	1.12	2.5	0.35
Embedded	3.6	1.20	2.5	0.32

Software Project Category: Semi Detached

Estimated Lines of Code (LOC): 3000 LOC = 3 KLOC

Effort applied = $a \cdot (\text{KLOC})^b$ [Person-Months] = $3.0(3)^{1.12}$ = 10.26 PM (Person-Months)

Development Time = $c \cdot (\text{Effort})^d$ = $2.5(10.26)^{0.35}$ = 5.64 months / 171 days

No: of Persons = Effort / Development Time = $10.26 / 5.64$ = 1.81 = 2 person

If 'X' is the salary of one person, then total development cost is 1X for this project.

3.8 Hardware Cost Estimation

Table 3.2 Hardware Cost Estimation

Sl. No.	Component Name	Count	Cost in INR
1	Raspberry Pi	1	8300
2	Raspberry Pi case and cooler	1	900
3	Web Camera HD 1080p	1	5000
4	Bread Board	1	55
5	Male to female connector	40	40
6	Male to male connector	40	41
7	ESP32	1	350
8	SIM800L 3.4V-4.4V	1	200
9	LED Bulb 5V, 200-500 lumens	1	50
10	Buzzer 3V	1	30
11	Lithium-ion Battery 3.7V 18650	2	200
12	Raspberry Pi Adapter	1	3500
13	Case for battery	1	18
Total amount in INR			18,884
Amount in words: Eighteen thousand and eight hundred eighty-four only			

3.9 Total Product Cost Estimation

The total product cost is the sum of both hardware cost and software cost. As we are not professional developers, the software cost can be considered as 0. So, the estimated cost for developing this product is:

$$\text{Total cost} = \text{Hardware cost} + \text{Software cost} = 18,884 + 0 = 18,884 \text{ INR}$$

3.10 Project Scheduling using Gantt chart

Figure 3.3: Project Schedule using Gantt chart

ID	TASK NAME	Start	Finish	Duration	2024		2025	
					Dec	Jan	Feb	Mar
1	Requirement Gathering	30/12/2024	07/01/2025	9d				
2	Literature Survey	30/12/2024	08/01/2025	10d				
3	Feasibility Analysis	09/01/2025	13/01/2025	5d				
4	System Modelling and Design	14/01/2025	27/01/2025	14d				
5	Hardware Implementation	27/01/2025	4/02/2025	9d				
6	ML Model Development	05/02/2025	24/02/2025	20d				
7	Android App Development	05/02/2025	24/02/2025	20d				
8	Unit Testing	05/02/2025	28/02/2025	23d				
9	Integration and Unit Testing	24/02/2025	8/03/2025	13d				
10	Deployment	09/03/2025	09/03/2025	1d				
11	Documentation	30/12/2024	09/03/2025	70d				

The Gantt chart shows the project scheduling. Gantt chart shows the start and finish date of the project. Our project phases will be completed as per the prescribed time schedule in the above Gantt chart shown in Figure 3.3. The starting date was 03/12/2024 and the project is expected to complete on 09/03/2025.

CHAPTER 4

METHODOLOGY

4.1 Proposed System

The proposed system integrates both hardware and machine learning algorithms to address the pressing issue of wildlife intrusion in agricultural landscapes. Central to our solution is the Raspberry Pi 5, a powerful single board computer, paired with a high-definition webcam such as the ASUS Webcam C3. This hardware setup forms the backbone of our animal detection system, enabling real-time capture and analysis of wildlife activity in agricultural fields.

At the heart of our system lies a custom machine learning model built upon TensorFlow Lite, trained on a diverse dataset of wild animals commonly found in agricultural regions. Our model achieves high accuracy in identifying species such as wild boars, elephants, tigers, and others, while minimizing false positives. This robust detection capability is essential for providing timely alerts to farmers and mitigating potential crop damage or livestock loss.

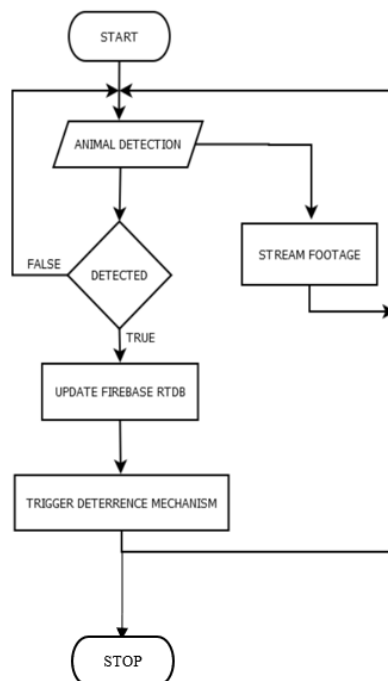


Figure 4.1: Flowchart of the system

In tandem with animal detection, our system incorporates humane deterrence methods to discourage wildlife intrusion without causing harm to animals or the environment. Through the ESP32 microcontroller, connected to a network of flashing lights and sound emitters, our system can activate deterrent measures in response to detected threats. This integrated approach ensures both the protection of agricultural assets and the promotion of ethical wildlife management practices.

To facilitate user interaction and system management, we have developed a companion Android app that provides live video feeds from the Raspberry Pi, enabling farmers to monitor their fields remotely. Additionally, the app allows for manual control of deterrence mechanisms and delivers real-time footage of detected wildlife activity. By empowering farmers with these tools and technologies, our system aims to enhance agricultural productivity while fostering harmonious coexistence between humans and wildlife in rural landscapes.

4.1.1 Animal Detection

Animal Detection consists of Raspberry Pi, camera, audio device and the machine learning model which detects wild animals. The custom ML model detects the wild animals from 7 classes and 1 class to avoid being detected. The classes to detect are Wild Boars, Elephants, Tigers, Bison, Foxes, Monkeys and Leopards. A class Human is also present in the custom dataset so that the model does not false identify a human. The system uses a Raspberry Pi-based board to have the camera and the ML model which identifies the animal. The ESP32 switch module will be controlled by the Raspberry Pi over the internet thus providing a wireless communication medium between both modules.



Figure 4.2 Examples for animals that are detected at night

4.1.2 Animal Deterrence

Deterrence techniques that have been utilized historically are quite risky; often, poison or explosives are used to keep wild animals away, which can be harmful to both the animal and the farmer in the case of an accident. Quickly flashing lights and loud sounds are used to deter the wild animals as these methods does not pose any harm to the life of the animal or the farmer. Animal Detection handles the identification and detection of the wild animals from the identification class, if the confidence of the detection of a prediction is above 70 percent the detection module sets the switch state to true, turning on the ESP 32 that controls the flashing of the lights.

4.2 Image Dataset

The dataset contains all labeled images of all the 8 classes that needs to be identified by the model. The images are collected from various sources from the internet and manually labelled, preprocessing and cleaning applied over the data set. The dataset can be approximately 3000 images in total from which 20 percent is taken for testing and another 20 percent for validation testing, the remaining data set goes fully to training the model. Each class represented in the dataset are: Wild Boar, Tigers, Elephants, Leopards, Monkeys, Foxes, Bison, and Humans.

The animals that can be detected with our system are:



Wild Boar



Elephant

Figure 4.3 Wild Animal Images



Leopard



Monkey



Bison



Tiger



Fox

Figure 4.3 Wild Animal Images

4.2.1 Training

The model chosen for the detection module is SSD MobileNet V2 FPNLite from the TensorFlow 2 Detection Model Zoo. The custom model is trained with training dataset. Google Colab's Pro GPU instance is used in training the model.

4.3 Device Management

An Android app is used for controlling the device manually. It displays the device's current condition and predictions. The app is built in android studio the Java programming language. The app feeds the video from the Raspberry Pi, providing a live feed for monitoring purposes. An audio device is to be used for producing sound when a wild animal is detected, which will be connected to the raspberry. The Android companion app functions provide live video streaming from the Raspberry Pi, allowing manual control of lights and sound, and displaying accurate device status and predictions.

4.4 Advantages of Proposed System

- Powerful and more efficient hardware.
- Live footage of detection in companion app.
- Manual control of deterrence mechanism.

CHAPTER 5

SYSTEM DESIGN

The proposed system model is depicted using a variety of system modelling techniques, including an architecture diagram, a use case diagram, an activity diagram, a sequence diagram, and a flow chart.

5.1 Architecture Diagram

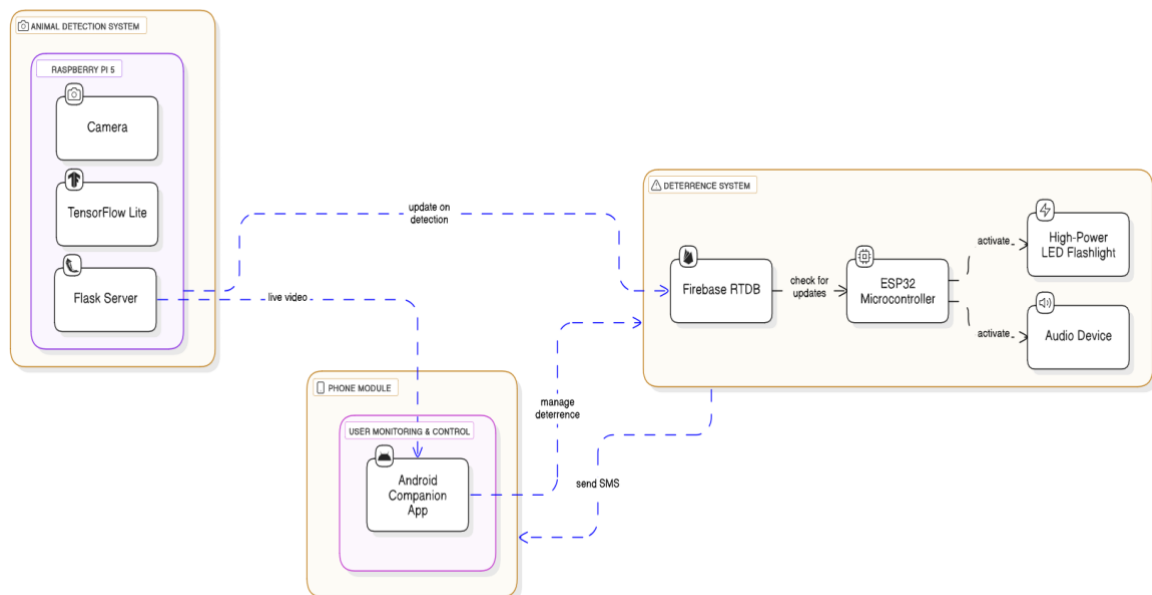


Figure 5.1 Architecture diagram of Animal Detection System

Description: -

The figure 5.1 illustrates a system using an ESP32 microcontroller to manage various components. The ESP32 communicates with a SIM800L GSM module for wireless communication. It also controls a relay module, which in turn controls an LED. Additionally, the ESP32 can activate a buzzer when needed. This setup allows for remote communication via GSM, controlling devices like LEDs through relays, and triggering alerts using a buzzer. The system can be useful for applications like remote monitoring, automation, and security alerts

5.2 Circuit Diagram

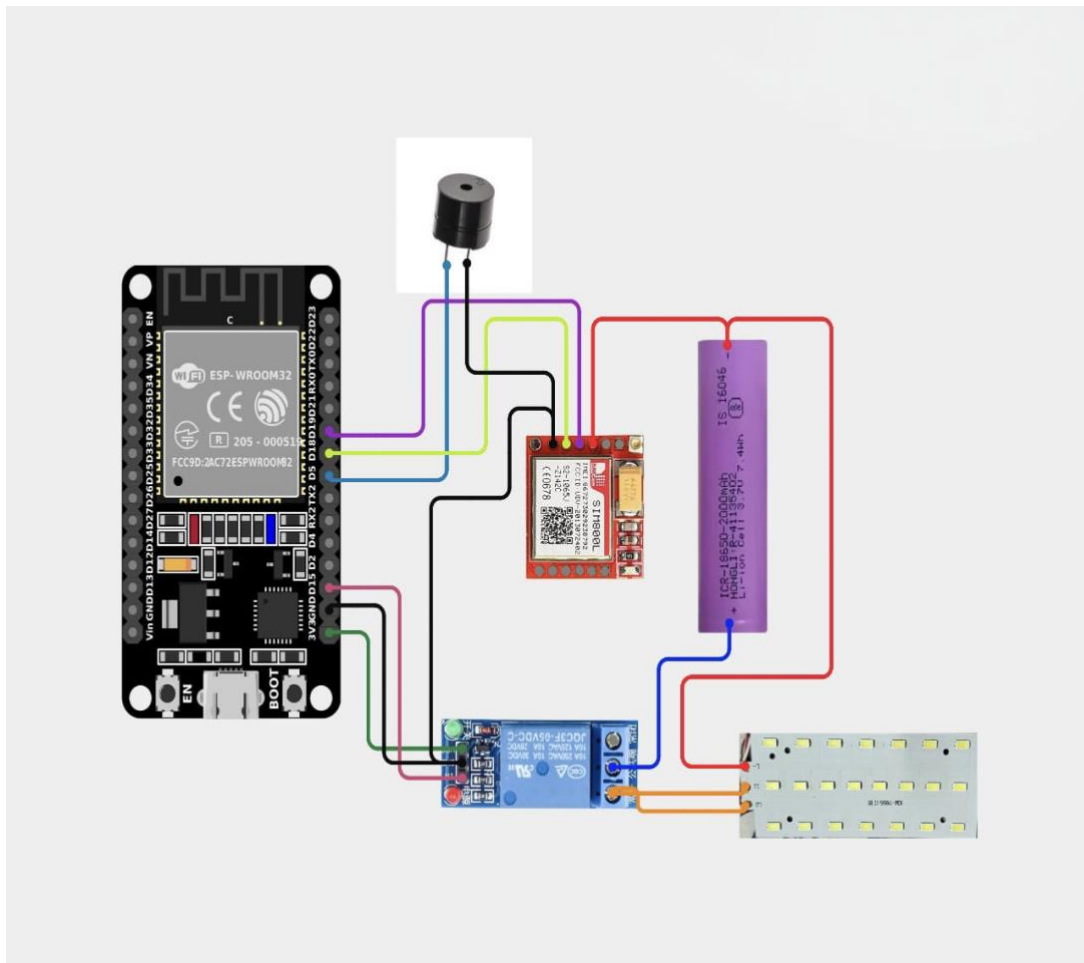


Figure 5.2 Circuit Diagram

Description: -

The figure 5.2 shows an ESP32 microcontroller connected to a SIM800L GSM module, a relay module, an LED strip, a buzzer, and a capacitor. The ESP32 communicates with the SIM800L for GSM-based functions, such as sending messages or receiving calls. The relay module is used to control the LED strip, allowing remote switching. The buzzer provides audible alerts. A capacitor is included to stabilize the power supply for the SIM800L, which requires steady voltage for reliable operation. This setup is ideal for IoT-based home automation, remote monitoring, or alert systems using GSM communication and ESP32 control.

5.3 Flow Chart

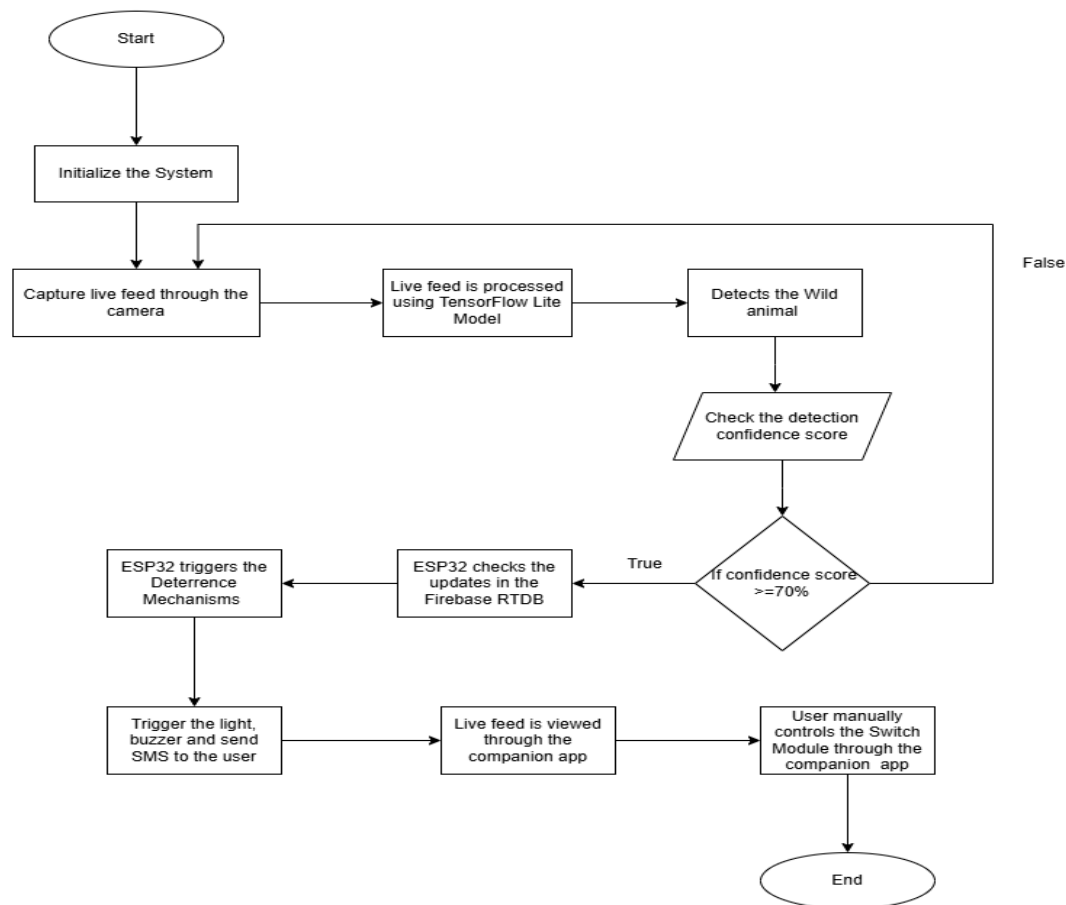


Figure 5.3 Flow Chart for Animal Intrusion Detection System

Description: -

The figure 5.3 shows a flow chart for Animal Intrusion Detection System. The system starts by continuously monitoring for animal presence using sensors or a camera. If no animal is detected, it loops back to continue monitoring. If an animal is detected, the system streams live footage for real-time monitoring and updates the Firebase Realtime Database (RTDB) for remote access. Simultaneously, a deterrence mechanism, such as an alarm, light, or sound, is triggered to scare the animal away. After executing the deterrence, the system either stops or resets to detect further activity, ensuring continuous surveillance and automated animal control for applications like farm security, wildlife deterrence, and smart monitoring.

5.4 Use Case Diagram

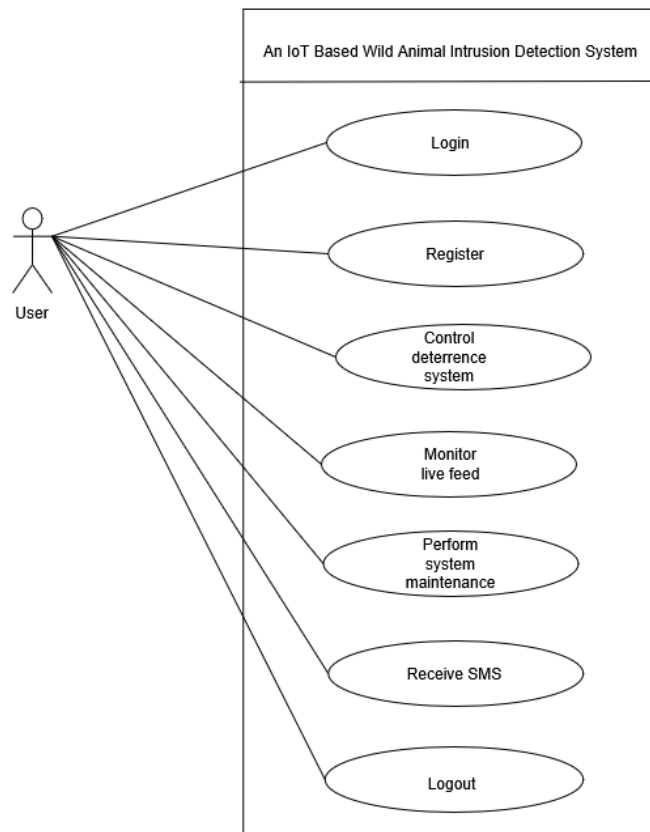


Figure 5.4 Use case diagram of the system

Description: -

The figure 5.4 shows use case diagram which represents the functionalities of the system, involving one actor: user. The user can log in/register, control the deterrence system, monitor the live feed, and logout. The user has all permissions including a function to perform system maintenance. The system allows users to monitor detected animals and take necessary actions using the deterrence mechanism. Users ensure smooth operation by maintaining the system. This diagram visually defines the system's scope, outlining interactions between users and the core functionalities of this system for effective wildlife monitoring and deterrence management.

5.5 Activity Diagram

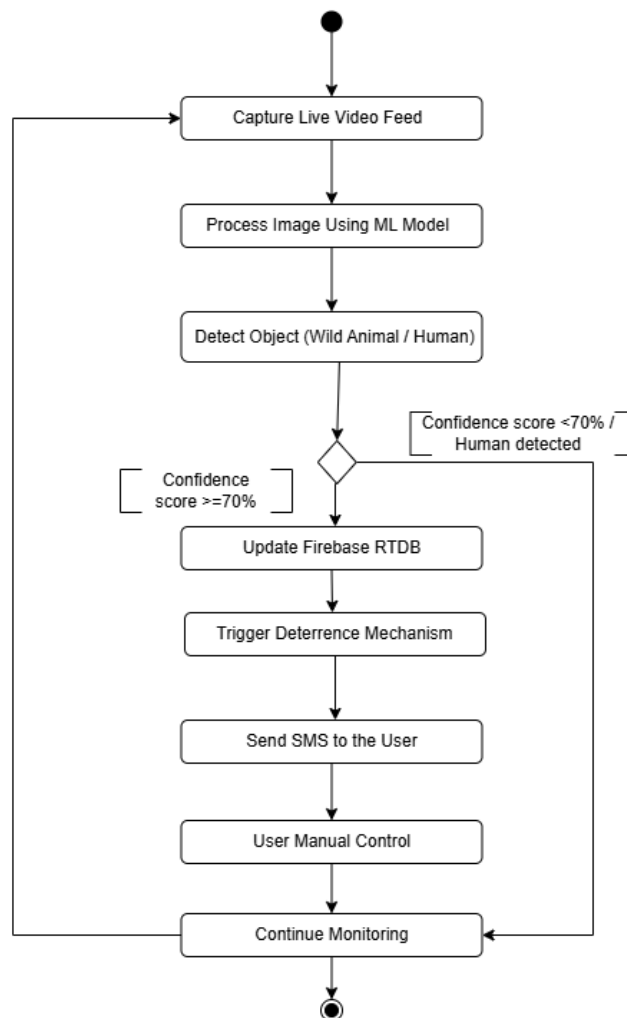


Figure 5.5 Activity diagram the system

Description: -

The figure 5.5 shows the flowchart of the system. It begins with detecting an object (Wild Animal/Human) and if detected, the system records live video of the object. The video is then processed by the ML Model (TensorFlow Lite). The system then checks the confidence score of the detected object (Wild Animal/Human). If the confidence score is below 70 percentage or the detected object is a human then the system will ignore the detected object and continue monitoring. If the confidence score is above or equal to 70 percentage then the updated data is sent to Firebase RTDB. The ESP32 then triggers the deterrence mechanism. An SMS is sent to the user. The user can manually control the live video via the companion app.

5.6 Sequence Diagram

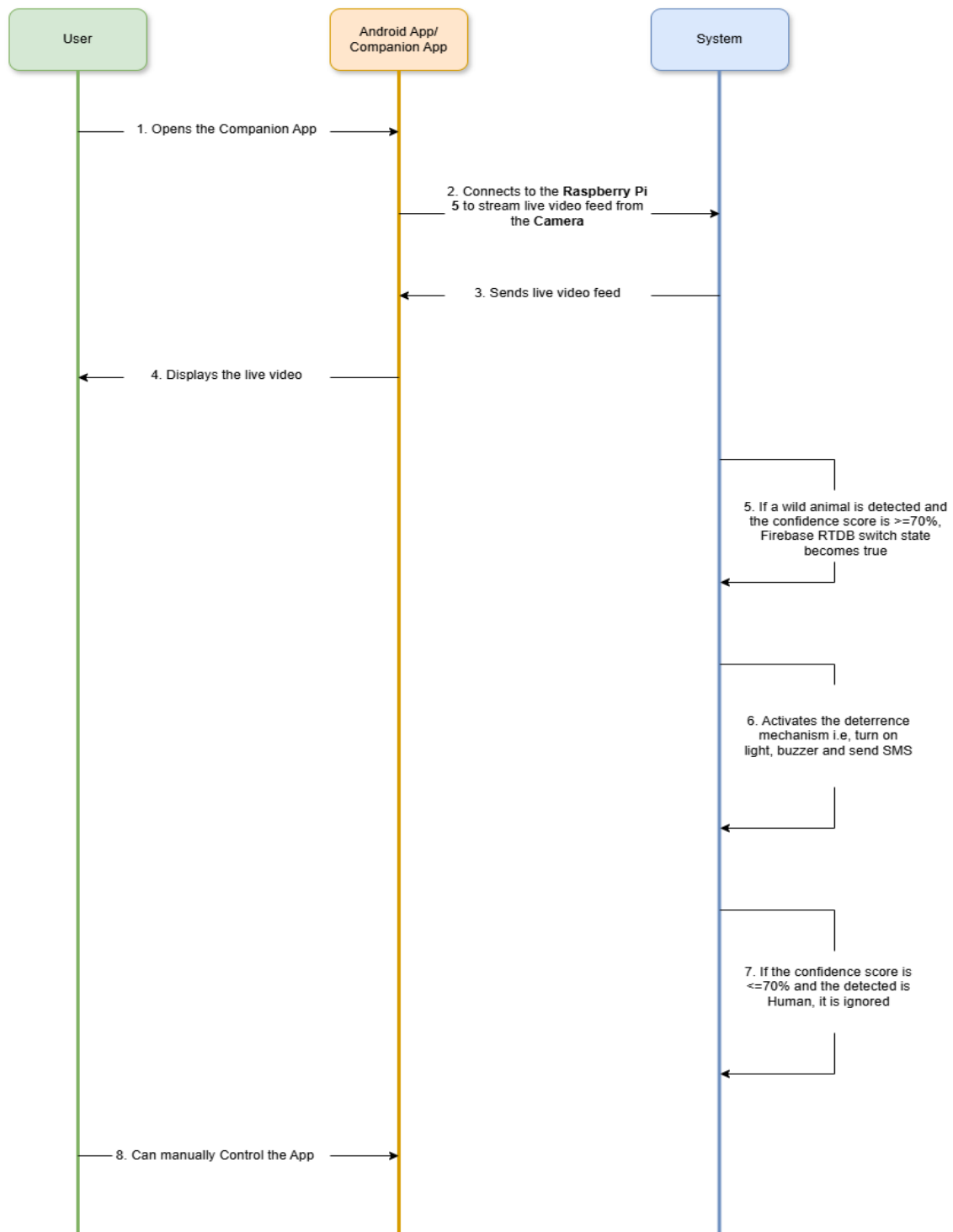


Figure 5.6 Sequence diagram the system

This sequence diagram illustrates the interaction flow between the User, the Android App/Companion App, and the System that is focused on wildlife detection and deterrent mechanisms. The system begins when the user initiates the process by opening the Companion App on their device, the App proceeds to establish a connection to a Raspberry Pi to access the camera. The Raspberry Pi streams the live video feed back to the Companion App, which then receives this feed. The App then displays the video to the user. The system monitors the video feed. If a wild animal is detected and the confidence level is equal to or greater than 70%, a switch in Firebase's Real-Time Database (RTDB) is activated. Upon validation of animal detection, the system activates deterrent measures by turning on a light, sounding a buzzer, and sending a SMS notification to the user. However, if the confidence score is less than or equal to 70% and the detected entity is identified as a human, the event is ignored. The user has the complete ability to manually control the app at any time during this process.

5.7 Collaboration Diagram

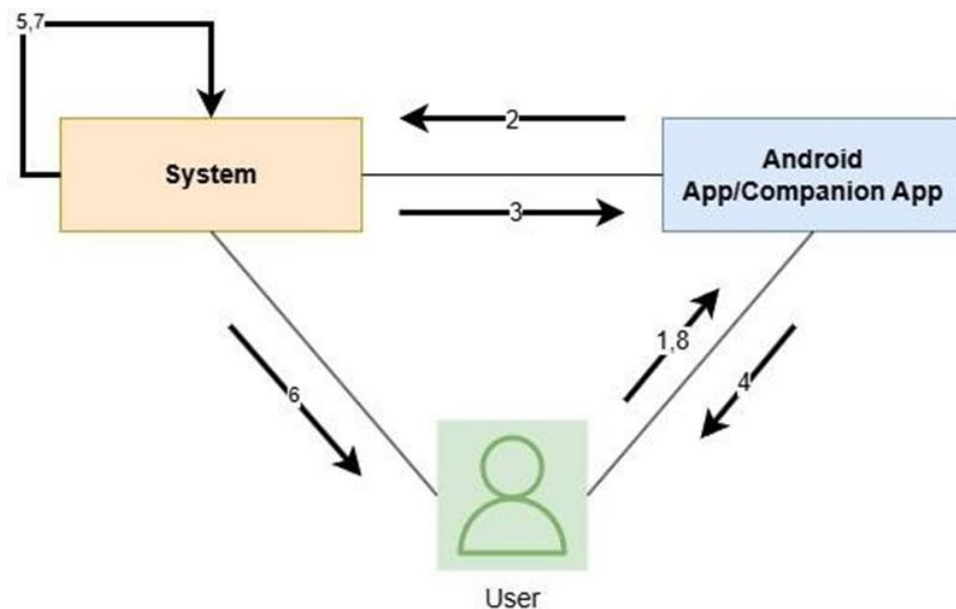


Figure 5.7 Collaboration diagram the system

The above collaboration diagram illustrates interactions between three main components: the User, the System, and the Android App/Companion App. The user interacts with both the System and the Android App. This interaction could include actions such as sending requests or receiving data. The System communicates bi-directionally with the Android App/Companion App. This communication typically involves data exchange, ensuring that the app has the latest information from the System. The user and the system interact when the system provides feedback to the user, possibly in response to user actions, then the user may provide information back to the system, fostering an interactive experience.

CHAPTER 6

SYSTEM IMPLEMENTATION

6.1 Machine Learning Model

In our project, machine learning is used to identify the different wild animals that can be intruded into the agricultural fields. The ML model is built using TensorFlow lite - SSD MobileNet V2 FPN lite is 320x320 is used to train the custom model.

TensorFlow lite is a special feature and mainly designed for embedded devices which is a solution for running machine learning models on mobile devices. The custom ML model detects seven kinds of wild animals while the class human is shielded from detection to prevent the model from mistakenly identifying a human. The classes to detect are Wild boars, Elephants, Tigers, Bison, Foxes, Monkeys and Leopards. The camera and the machine learning model that recognizes the animal are both built inside a Raspberry Pi based board. The raspberry pi will use the internet to control the ESP32 switch module, enabling wireless communication between two modules.

Steps involved in implementing the system are:

i) Wildlife Behavior Analysis: Perform a comprehensive evaluation of the agricultural site to observe and understand the behavior and habits of the local wildlife. Identify the primary pathways wild animals use to approach crops and pinpoint areas that may be susceptible to damage.

ii) Deterrence Method Selection: Using the insights gained from the wildlife behavior analysis, choose the most effective deterrence strategies suited to the specific species and their actions. Options may include methods such as flashing lights or loud noises to discourage wildlife.

iii) Strategic Deployment: Identify the best locations for placing deterrence devices, informed by the site assessment and wildlife behavior analysis. Install deterrents in areas where wildlife activity is most frequent or where crops are most vulnerable to ensure maximum effectiveness.

The dataset consists of 7 detected class and 1 undetected class, when the class names are initially added to the list when the dataset is loaded and it is stored in a jpg file which is converted to XML file after labelling. They are trained using a model built on the TensorFlow Lite architecture, and the result model is then translated to .tflite format. In this system we will create a dataset of wild animal images for the

classification of detected and undetected animals then we must fine-tune the model. The data should provide the model with some useful features. The Raspberry Pi runs a machine learning model built using TensorFlow Lite, specifically the SSD MobileNet V2 FPNLite architecture. The model is trained to identify various wild animals (e.g., wild boars, elephants, tigers) based on a dataset of labeled images. As the video feed is processed, the model analyzes each frame to detect animals. If an animal is detected with a confidence level above 70%, the model triggers the next steps. Upon detecting an animal, the Raspberry Pi updates the Firebase Realtime Database (RTDB) with the detected results, which includes the type of animal detected and the state of the switch which smooths out the real time communication between the Raspberry Pi and the ESP32 microcontroller. The ESP32 microcontroller continuously checks the Firebase RTDB for updates. When it detects a change in the switch state, it activates the deterrence mechanisms like flashing lights and emitting loud sounds to scare away the wild animals. The Android App allows users to monitor the system remotely. Users can view the live feed from the camera and is able to receive real time notifications, alerting users when an animal is detected.

6.2 Implementing the Model into Android App

Java language is used to construct android apps. The machine learning model is trained using TensorFlow and then converted to TensorFlow Lite format for efficient execution on edge devices like Raspberry Pi. The trained TensorFlow Lite model is hosted on the Raspberry Pi, which processes the video feed from the camera to detect animals in real-time. The Raspberry Pi runs a Flask server that serves the live video feed and handles requests from the Android app. The app and Raspberry Pi communicate through Firebase RTDB, which stores detection results and system status. The Android app does not execute the machine learning model directly. Instead, it relies on the Raspberry Pi to perform the detection tasks. The Android app uses a WebView to display the live feed from the Raspberry Pi which allows the users to monitor the field in real-time through the app. It updates the Firebase RTDB with the detection results. The android app provides for users to manually activate and deactivate the deterrence mechanisms using buttons. When a user presses a button, the app sends a command to update the corresponding values in Firebase RTDB.

6.3 Hardware Implementation

In the hardware implementation the major hardware components used are: -

6.3.1 Raspberry Pi 5

The Raspberry Pi 5, launched in early 2024, is a significant advancement in the single-board computer series, designed for a variety of applications including robotics, home automation, media centers, and educational tools. It is powered by a 2.4 GHz quad-core 64-bit ARM Cortex-A76 processor, offering 2-3 times the performance of its predecessor, the Raspberry Pi 4. This results in smoother multitasking and faster application loading times. Memory is available in 4GB or 8GB LPDDR4X SDRAM configurations, the Raspberry Pi 5 ensures efficient operation even with multiple applications running simultaneously. It utilizes a microSD card for storage, allowing for easy expandability and customization based on project needs. It also features an 800MHz VideoCore VII GPU for enhanced graphics performance, supporting high-resolution video playback and light gaming. It includes dual HDMI ports for connecting two displays at once. The redesigned Image Signal Processor enhances camera integration, making it suitable for computer vision projects. Users may need to adjust some applications for compatibility with the new hardware.



Figure 6.1 Raspberry Pi 5

6.3.2 ESP 32

The ESP32, developed by Espressif Systems, is a versatile microcontroller and system-on-chip (SoC) widely utilized in Internet of Things (IoT) and embedded systems applications. The ESP32 operates at speeds up to 240 MHz, providing robust processing capabilities for connected devices. It has Integrated Wi-Fi and Bluetooth support for both 2.4 GHz and 5 GHz bands, making it suitable for various wireless communication applications, from simple IoT sensors to complex smart home devices. Additionally, it offers a rich set of peripherals, including GPIO pins, SPI, I2C, I2S, UART, CAN, Ethernet MAC, SDIO/SPI, PWM, ADC, and DAC, allowing

seamless integration with sensors, actuators, and displays. It also features multiple sleep modes and power-saving options, making it ideal for battery-powered and energy-efficient applications. Moreover, supports external SPIRAM for additional memory, enabling the development of memory-intensive applications like audio processing and machine learning. ESP32 provides robust development tools and software frameworks, including Arduino IDE, ESP-IDF, and MicroPython, facilitating rapid prototyping and development and with its compact size, low cost, and extensive feature set, the ESP32 has become a popular choice among hobbyists, makers, and professional developers for a wide range of projects.



Figure 6.2 ESP 32

6.3.3 SIM 800L

The SIM800L is a GSM/GPRS module that is often used in IoT applications for enabling mobile communication. In the context of this project, it can be utilized for sending alerts or notifications regarding animal detections to users via SMS or for other communication purposes. The SIM800L is a quad-band GSM/GPRS module, which means it can operate on multiple frequency bands, making it suitable for global use. The SIM800L operates at a voltage range of 3.4V to 4.4V, with an optimal voltage of 4.0V. The module can send and receive SMS messages, making it useful for alerting users about animal detections. The SIM800L is small and lightweight, making it suitable for integration into various projects without adding significant bulk.



Figure 6.3 SIM 800L

6.3.4 Relay Module

A 3V to 5V single-channel relay is a type of electromechanical switch that is designed to be activated by a low voltage signal, typically between 3V and 5V. This relay typically consists of a coil and a set of contacts. When the coil is energized with the specified voltage (3V to 5V in this case), it creates a magnetic field that causes the contacts to close or open, depending on the relay type.



Figure 6.4 Relay Module

6.3.5 Web Camera

The camera used is normal webcam that has the ability to capture up to 1080p footage. The specific model used in this work is ASUS Webcam C3, it is a USB camera with 1080p 30 fps recording. The PI camera module can also be used with this work with a little bit of alteration to the to the detection code so that it captures the footage from the PI camera module instead of the USB webcam



Fig 6.5 Web Camera

6.3.6 Buzzer

The project typically employs an active buzzer, which generates sound when a voltage is applied. Active buzzers contain an internal oscillator, making them easier to use since they only require a DC voltage to produce sound. The buzzer operates within a voltage range of 3V to 5V, making it compatible with the power supply used in this system. The sound output of the buzzer can vary, typically producing sound levels around 85 dB at a distance of 1 meter, which is loud enough to deter animals effectively. The frequency of the sound produced by the buzzer is usually in the range of 2 kHz to 4 kHz, which is effective for startling animals. The current consumption of the buzzer is generally low, often around 20 mA to 30 mA when activated, allowing it to operate efficiently without draining the battery quickly.



Fig 6.6 Buzzer

6.3.7 LED Light

The project typically uses high-power LED strips or individual LED bulbs that can produce bright flashes of light. The LED lights are powered by a 3.7V battery, ensuring that they can operate effectively in remote agricultural settings where traditional power sources may not be available. These LEDs are chosen for their brightness and effectiveness in deterring wild animals. The brightness of the LEDs can range from 100 lumens to over 1000 lumens per meter for LED strips, providing sufficient illumination to deter animals effectively.



Fig 6.7 LED Light

6.3.8 Battery

In this system we use two 3.7V lithium-ion (Li-ion) batteries which is typically used to power the LED lights and other components of the system. This type of rechargeable battery is favored for its high energy density, lightweight, and ability to provide a stable voltage output. The battery operates at a nominal voltage of 3.7V, which is suitable for powering the LED lights and the ESP32 microcontroller. Li-ion batteries typically have a discharge rate of 1C to 2C, meaning they can safely discharge their capacity at a rate equal to or double their capacity.



Fig 6.8 Battery

CHAPTER 7

TESTING

7.1 Testing

Testing is the process used to evaluate a software product's usability, efficiency, and quality to make sure that it complies with the criteria and specifications laid down for it in the frame of software development. In order to find bugs, failures, or other problems that can impair the software's operation, usability, or performance, testing often entails running the program under a variety of circumstances and input circumstances. Testing is a crucial step in the development of software because it enables the detection and correction of any problems or flaws before the product is made available to consumers. Unit testing, integration testing, system testing, acceptance testing, and regression testing are just a few of the several forms of testing that may be done. Each kind of testing has a particular function and is created to handle particular faces of the software development process.

7.1.1 Types of Testing

Various types of testing can be carried out while developing software. Here are a few of the most typical:

Unit Testing: - This is the process of checking that individual software program units or components function as planned.

Integration Testing: - This entails evaluating how various parts or units of a software program interact with one another to make sure they are compatible and perform as intended.

System Testing: -This entails testing the entire system to make sure it complies with the guidelines and standards established for it.

Acceptance Testing: - In preparation for a software product being made available to users, this is the last phase of testing. To make sure the program satisfies the user's needs, it must be tested in a setting similar to a production one.

Regression Testing: -Retesting a software program after modifications have been made is necessary to make sure that no new defects or problems were created by the changes.

Performance Testing: -This entails testing the software to make sure it operates effectively under real-world or hypothetical pressures.

Security Testing: -Testing the software is necessary to find any flaws or vulnerabilities that an attacker could use against it.

Usability Testing: -This entails evaluating the software to make sure that its intended users would find it simple to use and understand.

Compatibility Testing: -To make sure it works in all intended environments, this requires testing the software program on various hardware, software, and operating system configurations.

Exploratory Testing: -This entails testing the software application without following a predetermined test strategy with the intention of identifying fresh and unexpected flaws.

7.1.2 Test Summary Report

Table 7.1 Decision Based Testing

MODULE UNDER TEST: Hardware Module – Animal Detection		RULES			
CONDITIONS	When an animal is detected by the detection module	T	T	F	F
	When the accuracy of the detection is ≥ 70	T	F	T	F
ACTIONS	Turn ON Light & Buzzer	✓			
	Turn OFF Light & Buzzer		✓	✓	✓
	Send SMS	✓			
	If no animal is detected, no SMS will be sent		✓	✓	✓

Table 7.2 Testing Summary of IoT Module

Test case Id	Function Under Test	Input	Expected Output	Actual Output	Pass/Fail
TC-01	Detection Module	Animal: Wild Boar	Light & Buzzer ON, Sent SMS	Light & Buzzer ON, Sent SMS	Pass
TC-02	Detection Module	Animal: Elephant	Light & Buzzer ON, Sent SMS	Light & Buzzer ON, Sent SMS	Pass
TC-03	Detection Module	Animal: Leopard	Light & Buzzer ON, Sent SMS	Light & Buzzer ON, Sent SMS	Pass
TC-04	Detection Module	Animal: Monkey	Light & Buzzer ON, Sent SMS	Light & Buzzer ON, Sent SMS	Pass
TC-05	Detection Module	Animal: Bison	Light & Buzzer ON, Sent SMS	Light & Buzzer ON, Sent SMS	Pass
TC-06	Detection Module	Animal: Tiger	Light & Buzzer ON, Sent SMS	Light & Buzzer ON, Sent SMS	Pass
TC-07	Detection Module	Animal: Fox	Light & Buzzer ON, Sent SMS	Light & Buzzer ON, Sent SMS	Pass
TC-08	Detection Module	Human	Light & Buzzer OFF, No SMS Sent	Light & Buzzer OFF, No SMS Sent	Pass

CHAPTER 8

RESULTS

We have built an Android application and an Animal Detection System which consists of two modules IoT module and ML module. The ML module consists of a web camera and a Raspberry Pi 5. The IoT module consists of an ESP 32, a Relay module, a GSM module (SIM 800L), buzzer, LED light and a battery to power GSM module and LED. When the ML module detects the wild animal it sends a signal to the IoT trigger a loud noise and bright light to scare the animal away. The Android app helps in viewing the detected animals with its specified class and accuracy value. The android app is used for controlling the device manually and it also provides the current status and predictions provided by the device.

8.1 Software Module

When the app is launched the user can see the pages listed below:

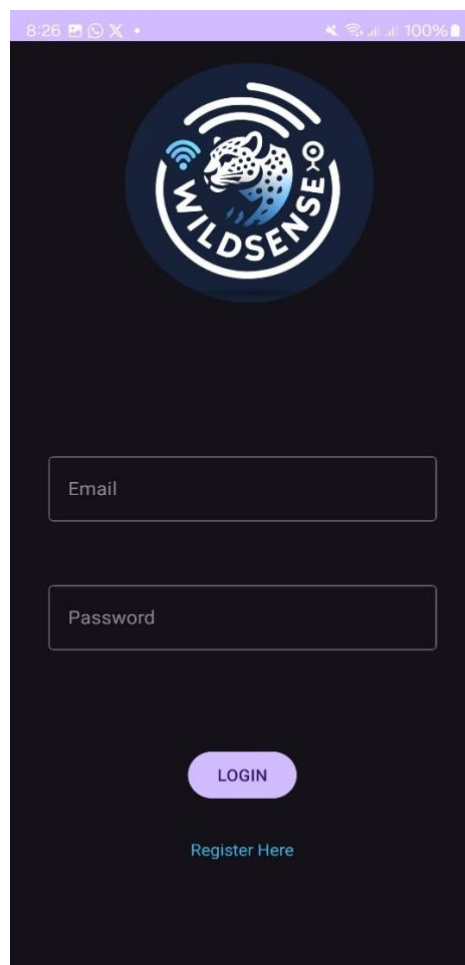


Figure 8.1 Start Screen

Description: -

The Figure 8.1 shows the starts screen of the Android app (WildSense). When the user opens the Android app there are three buttons ON and OFF to control the buzzer and light manually and REFRESH STREAM button to refresh the app. While no animal is being detected the user can control the light and buzzer using the ON and OFF buttons. It can also be used to ensure that the system is working properly.

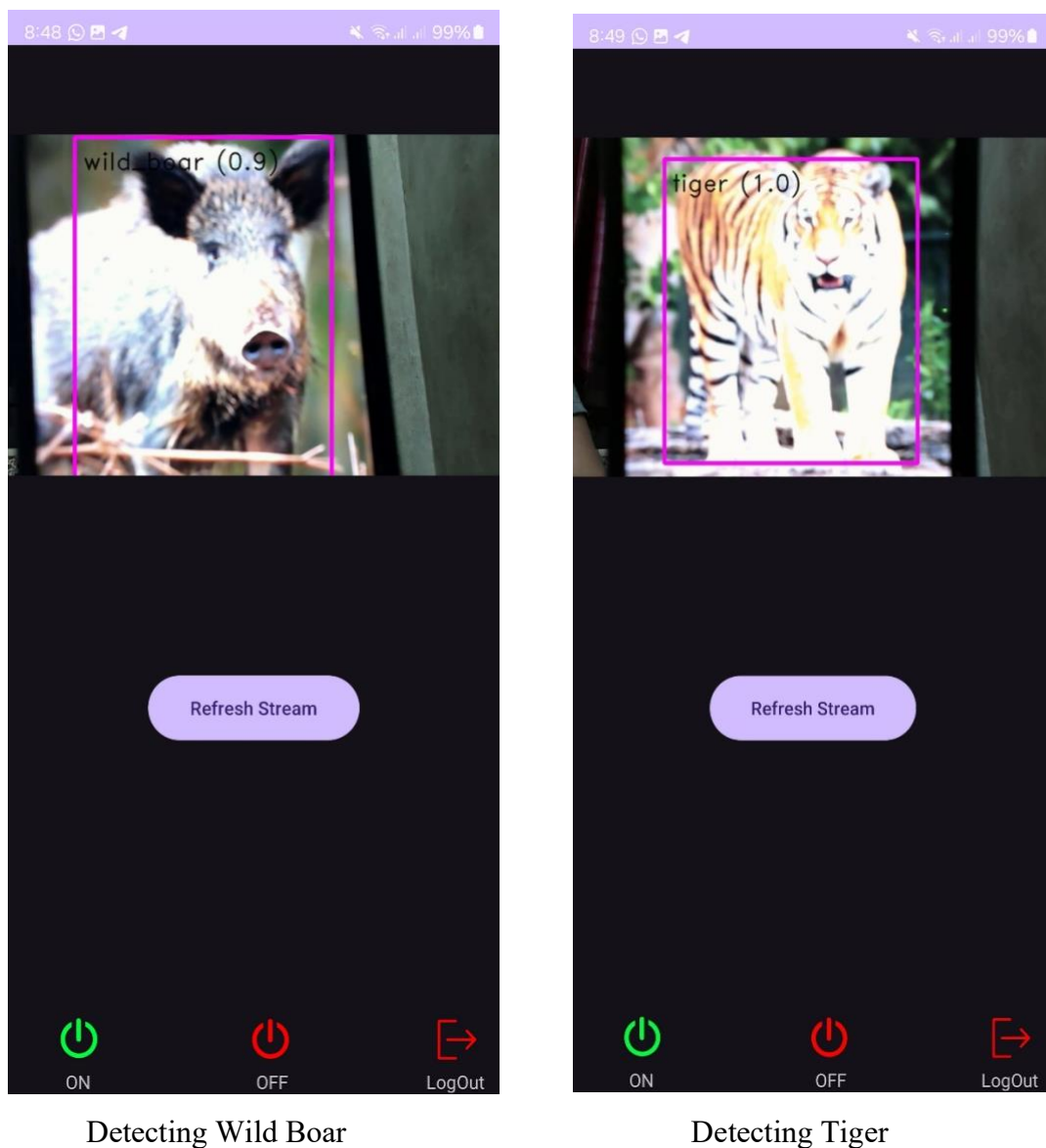
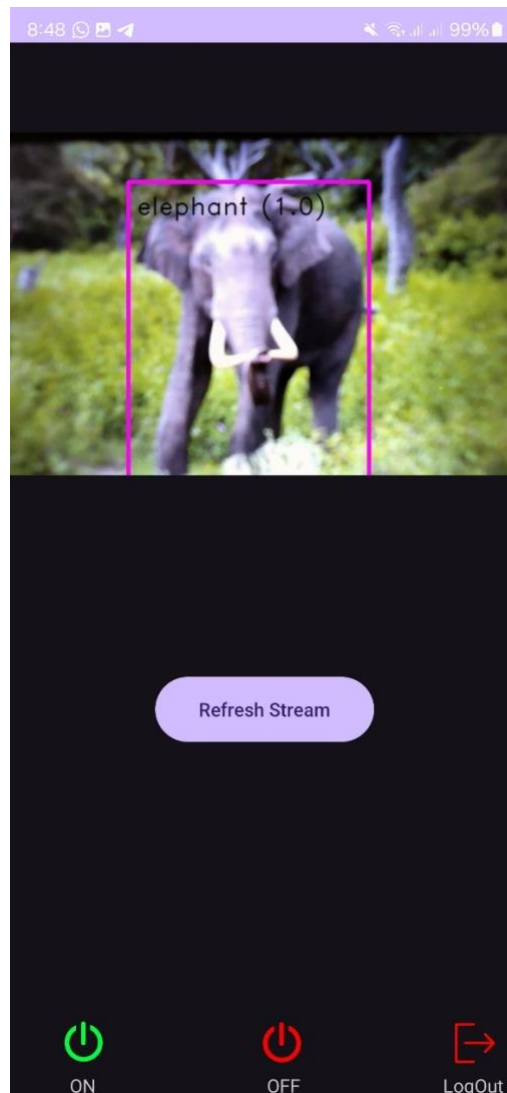


Figure 8.2 Detecting Wild Animals



Detecting Elephant

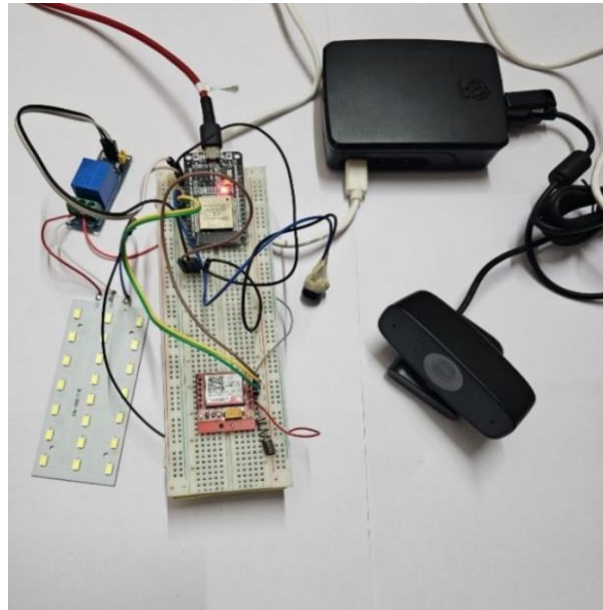
Fig 8.2 Detecting wild animals

Description: -

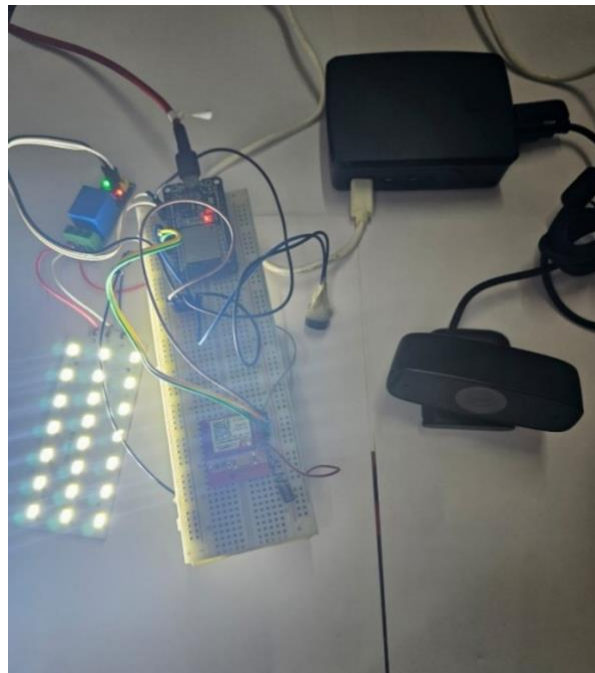
After the animal is being detected the live footage can be viewed by the user in the Android app WildSense. The class and accuracy value of the corresponding animal is also shown in a bounding box. The accuracy values in the Fig 8.2 are 0.98 for wild boar, 0.99 for tiger and 0.99 for elephant. This indicates that the images are 98%, 99%, 99% accurate that it is the respective animal.

If the app is not showing anything or if there is any connection issues the user can click on the “Refresh Stream” button which reloads the page.

8.2 Hardware Module



While not detecting



While detecting

Figure 8.3 Animal Detection System

In Figure 8.3, when the ML module detects a wild animal with 70% accuracy it sets the switch state as TRUE for the switch module in Firebase RTBD. This activates the ESP 32, which in turn activates the loud noise and light to scare the animals away

CHAPTER 9

CONCLUSIONS

In Kerala the number of wild animal attacks over farmlands and residential areas have increased over the past few years. Our project aims to provide a mechanism to reduce these incidents without causing any harm to both the user of this system and the animal. Our project is a complete and efficient way to mitigate wild animal risks in agricultural areas. Our project uses hardware devices like surveillance cameras and deterrence mechanisms, as well as software and deployment tactics, to monitor wildlife behavior and respond to possible hazards. The project's goal is to scare off the animals from the field without endangering them, saving the farmer's as well as the animal's life. The camera and machine learning model that recognizes the animal are installed on a Raspberry Pi-based board in this project. The Raspberry Pi will be able to control the ESP32 switch module via the internet, enabling wireless communication between the two modules. The Raspberry Pi's live feed is streamed live via the Android companion app, which also allows the user to take manual control of the deterrence mechanism.

REFERENCES

- [1] J. Miao, D. Rajasekhar, S. Mishra, S. K. Nayak and R. Yadav, "A Fog-based Smart Agriculture System to Detect Animal Intrusion," 2023 IEEE 29th International Conference on Parallel and Distributed Systems (ICPADS), Ocean Flower Island, China, 2023, pp. 2523-2530, doi: 10.1109/ICPADS60453.2023.00336.
- [2] S. Jeevitha and S. V. Kumar, "A Study on Sensor Based Animal Intrusion Alert System Using Image Processing Techniques," 2019 Third International conference on I-SMAC (IoT in Social, Mobile, Analytics and Cloud) (I-SMAC), Palladam, India, 2019, pp. 20-23, doi: 10.1109/I-SMAC47947.2019.9032430.
- [3] B. K and S. J, "Animal Detection and Prevention in Agri Field Using IoT *," 2022 Third International Conference on Smart Technologies in Computing, Electrical and Electronics (ICSTCEE), Bengaluru, India, 2022, pp. 1-6, doi: 10.1109/ICSTCEE56972.2022.10099831.
- [4] B. Meenakshi, A. B. SR Y, H. N C, K. L and A. J, "Animal Intrusion Detection and Ranging system using YOLOv4 and LoRa," 2022 International Conference on Power, Energy, Control and Transmission Systems (ICPECTS), Chennai, India, 2022, pp. 1-6, doi: 10.1109/ICPECTS56089.2022.10047729.
- [5] K. M, B. V and A. K, "Animal Intrusion Detection Using Yolo V8," 2024 10th International Conference on Advanced Computing and Communication Systems (ICACCS), Coimbatore, India, 2024, pp. 206-211, doi: 10.1109/ICACCS60874.2024.10716895.
- [6] A. P E, A. C K, M. Shibili, R. C K and N. K, "Development and Implementation of an Animal Intrusion Detection System Using Image and Audio Processing," 2023 14th International Conference on Computing Communication and Networking Technologies (ICCCNT), Delhi, India, 2023, pp. 1-7, doi: 10.1109/ICCCNT56998.2023.10307574.
- [7] P. K. Panda, C. S. Kumar, B. S. Vivek, M. Balachandra and S. K. Dargar, "Implementation of a Wild Animal Intrusion Detection Model Based on Internet of Things," 2022 Second International Conference on Artificial Intelligence and Smart Energy (ICAIS), Coimbatore, India, 2022, pp. 1256-1261, doi: 10.1109/ICAIS53314.2022.9742948.
- [8] M. Begum H., D. A. Janeera and A. Kumar. A.G, "Internet of Things based Wild Animal Infringement Identification, Diversion and Alert System," 2020 International Conference on Inventive Computation Technologies (ICICT), Coimbatore, India, 2020, pp. 801-805, doi: 10.1109/ICICT48043.2020.9112433.
- [9] T. Manikandan, S. J. Kumaresan, A. Muruganandham, V. Nandalal and R. Babuji, "IoT Based Animal Detection and Alert System for Farm Fields," 2024 International Conference on Communication, Computing and Internet of Things (IC3IoT), Chennai, India, 2024, pp. 1-3, doi: 10.1109/IC3IoT60841.2024.10550250.

- [10] S. Usharani, Gayathri, R. S, D. S. Kishore and S. Depuru, “IoT based Animal Trespass Identification and Prevention System for Smart Agriculture,” 2023 7th International Conference on Intelligent Computing and Control Systems (ICICCS), Madurai, India, 2023, pp. 983-990, doi: 10.1109/ICICCS56967.2023.10142814.
- [11] S. Priyanka, S. A. Suje, D. Selvapandian and V. Narasimharaj, “Smart Farming: Intelligent Animal Detection Framework in Agriculture using IoT Sensors,” 2023 2nd International Conference on Edge Computing and Applications (ICECAA), Namakkal, India, 2023, pp. 1336-1341, doi: 10.1109/ICECAA58104.2023.10212297.
- [12] Prof. Divya, Prof. Usha kiran and Prof. Praveen M IOT-Based Wild Animal Intrusion Detection System 2018 <http://www.ijritcc.org>

APPENDIX 1: SAMPLE CODE

```
#include <Arduino.h>

#if defined(ESP32) || defined(ARDUINO_RASPBERRY_PI_PICO_W)

#include <WiFi.h>

#elif defined(ESP8266)

#include <ESP8266WiFi.h>

#endif

#include <Firebase_ESP_Client.h>

#include <addons/TokenHelper.h>

#include <addons/RTDBHelper.h>

#define WIFI_SSID "WildSense"

#define WIFI_PASSWORD "1234"

#define API_KEY "AIzaSyD8L3KvO9XXmLw5uo2PhE5xM5V0a3xGhRI"

#define DATABASE_URL "https://wildsense-f4d32-default-rtdb.asia-southeast1.firebaseio.com/"

#define rxPin 18

#define txPin 19

HardwareSerial sim800(1);

String PHONE = "+919567383456";

FirebaseData fbdo;

FirebaseAuth auth;

FirebaseConfig config;

bool signupOK = false;

const int buzzer = 5;
```

```
const int LEDPIN = 15;

unsigned long previousMillis = 0;
unsigned long alertMillis = 0;
const long interval = 3000;
const long alertInterval = 120000;

bool switchval;
String label;

void setup()
{
  Serial.begin(115200);
  Serial.println("esp32 serial initialize");
  sim800.begin(9600, SERIAL_8N1, rxPin, txPin);
  Serial.println("SIM800L serial initialize");
  sim800.println("AT+CMGF=1"); // SMS text mode
  delay(1000);
  sim800.println("AT+CMGD=1,4"); // Delete all saved SMS
  delay(1000);

  pinMode(buzzer, OUTPUT);
  pinMode(LEDPIN, OUTPUT);
  digitalWrite(LEDPIN, HIGH);

  WiFi.begin(WIFI_SSID, WIFI_PASSWORD);

  while (WiFi.status() != WL_CONNECTED)
  {
    Serial.println("Connecting...");
    delay(300);
  }
}
```

```
}

config.api_key = API_KEY;
config.database_url = DATABASE_URL;

if (Firebase.signUp(&config, &auth, "", ""))
{
    signupOK = true;
    Serial.println("Firebase Connected");
}

config.token_status_callback = tokenStatusCallback;
Firebase.begin(&config, &auth);
Firebase.reconnectWiFi(true);
}

void loop()
{
    unsigned long currentMillis = millis();

    if (currentMillis - previousMillis >= interval)
    {
        previousMillis = currentMillis;

        if (Firebase.RTDB.getBool(&fbdo, F("detected_classes/switch")))
        {
            switchval = fbdo.boolData();
            Serial.println(switchval);
        }
    }
}
```

```
if (Firebase.RTDB.getString(&fbdo, F("detected_classes/label")))  
{  
    label = fbdo.stringData();  
}  
}
```

```
if (switchval)  
{  
    Serial.println("xdd");  
    if (alertMillis == 0 || currentMillis - alertMillis >= alertInterval)  
    {  
        sendAlert(label);  
        alertMillis = currentMillis;  
    }  
}
```

```
digitalWrite(LEDPIN, LOW);
```

```
for (unsigned char i = 0; i < 80; i++)  
{  
    digitalWrite(buzzer, HIGH);  
    delay(1);  
    digitalWrite(buzzer, LOW);  
    delay(1);  
}
```

```
for (unsigned char i = 0; i < 100; i++)  
{  
    digitalWrite(buzzer, HIGH);  
    delay(2);  
    digitalWrite(buzzer, LOW);  
}
```



```
        delay(2);
    }
}
else
{
    digitalWrite(LEDPIN, HIGH);
}

    delay(10);
}

void sendAlert(String animal)
{
    String sms_data;
    sms_data = "Wild Animal Detected: " + animal;

    sim800.print("AT+CMGF=1\r");
    delay(1000);
    sim800.print("AT+CMGS=\"" + PHONE + "\"\r");
    delay(1000);
    sim800.print(sms_data);
    delay(100);
    sim800.write(0x1A);
    delay(1000);
    Serial.println("SMS Sent Successfully.");
    delay(1000);
}
```

APPENDIX 2: BILLS OF PURCHASED MATERIALS



INVOICE

Invoice No: #9088 Date: 20 Jan 2025

Invoice To:

 Abiya Mariyam Alex,
 Vadakkadathu House, Muttom P.O Thumpamon, 689502

Pay To:

 Robocraze, Survey no. 104, Ground Floor, 912/10, 4th G Main Rd, Chellikere, Chelekare,
 Kalyan Nagar, Bengaluru, Karnataka 560043

59

Description	Price	Qty	CGST	SGST	Total
ESP32 Node MCU	347	1	0 %	0 %	₹ 347
Development Board with Wifi and Bluetooth (CP2102 Driver, 30 PIN)			₹ 0.00	₹ 0.00	
Male to Female Jumper	39	1	0 %	0 %	₹ 39
Wires (20 cm) 40pcs			₹ 0.00	₹ 0.00	
Male to Male Jumper	41	1	0 %	0 %	₹ 41
Wires (20 cm) 40pcs			₹ 0.00	₹ 0.00	
18650 Single Cell Battery	18	1	0 %	0 %	₹ 18
Case Holder			₹ 0.00	₹ 0.00	
Sim800L GSM GPRS Module	202	1	0 %	0 %	₹ 202
			₹ 0.00	₹ 0.00	
1 CH 5V Active Low Relay Board	32	1	0 %	0 %	₹ 32
			₹ 0.00	₹ 0.00	
18650 2200mAh 3.7v Cylinder Battery	142	1	0 %	0 %	₹ 142
			₹ 0.00	₹ 0.00	
Payment info:			Subtotal		₹ 821
Online					
Amount: ₹ 821					

Terms & Conditions:

This is an invoice for confirmation of the receipt of the amount paid against for the service as described above. * Keep this invoice and manufacturer box for warranty purpose .